MORPHEUS model construction for semiexperimental structures: coordinates, constraints, classes, and predicates

MATRIX/MORPHEUS working draft

July 1, 2026

Abstract

Semiexperimental equilibrium structures are often limited less by the least-squares algorithm than by the definition of the refinement model. MORPHEUS treats model construction as an explicit, auditable step. The user may refine in symmetry-adapted generalized internal coordinates (GICs) or in symmetry-adapted Cartesian coordinates, while chemical knowledge is always expressed at the primitive internal-coordinate level. This draft summarizes the available model choices: fully free refinements, hard constraints, automatically generated parameter classes, chemically defined primitive classes, and weighted predicates. Initial tests on fructose and deoxy-D-ribose show that GIC and Cartesian active spaces give numerically equivalent results once the model is well posed, whereas GICs provide the natural language for chemical constraints, diagnostics, and published structural parameters.

1 Purpose

The central design choice in MORPHEUS is to separate the numerical active space from the chemical model. Rotational constants or moments of inertia are fitted in a well-defined coordinate basis, but all structural assumptions are expressed as primitive distances, valence angles, torsions, out-of-plane coordinates, ring coordinates, or their symmetry-adapted combinations. This makes each refinement reproducible: the input geometry, isotopic data, vibrational corrections, electronic corrections, constraints, predicates, parameter classes, rank, covariance, and warnings are all written to the run manifest and text report.

2 Coordinate models

2.1 Symmetry-adapted GICs

The default MORPHEUS active space is the non-redundant totally symmetric block of the GIC set generated by NEO/GICForge. Primitive coordinates are generated from topology, reduced to a non-redundant set, symmetry-adapted, and labelled. The active coordinates include stretches, bends, torsions, out-of-plane coordinates, ring puckering coordinates, and special coordinates when present. The GIC route has three advantages:

- chemical constraints can be projected directly from primitive coordinates;
- parameter classes can be defined in chemically meaningful families;
- propagated errors can be reported for bond lengths, angles, torsions, and ring puckering coordinates.

2.2 Symmetry-adapted Cartesians

The alternative active space is a symmetry-adapted Cartesian displacement basis after removal of translations and rotations. This route is useful as an independent numerical check because it avoids

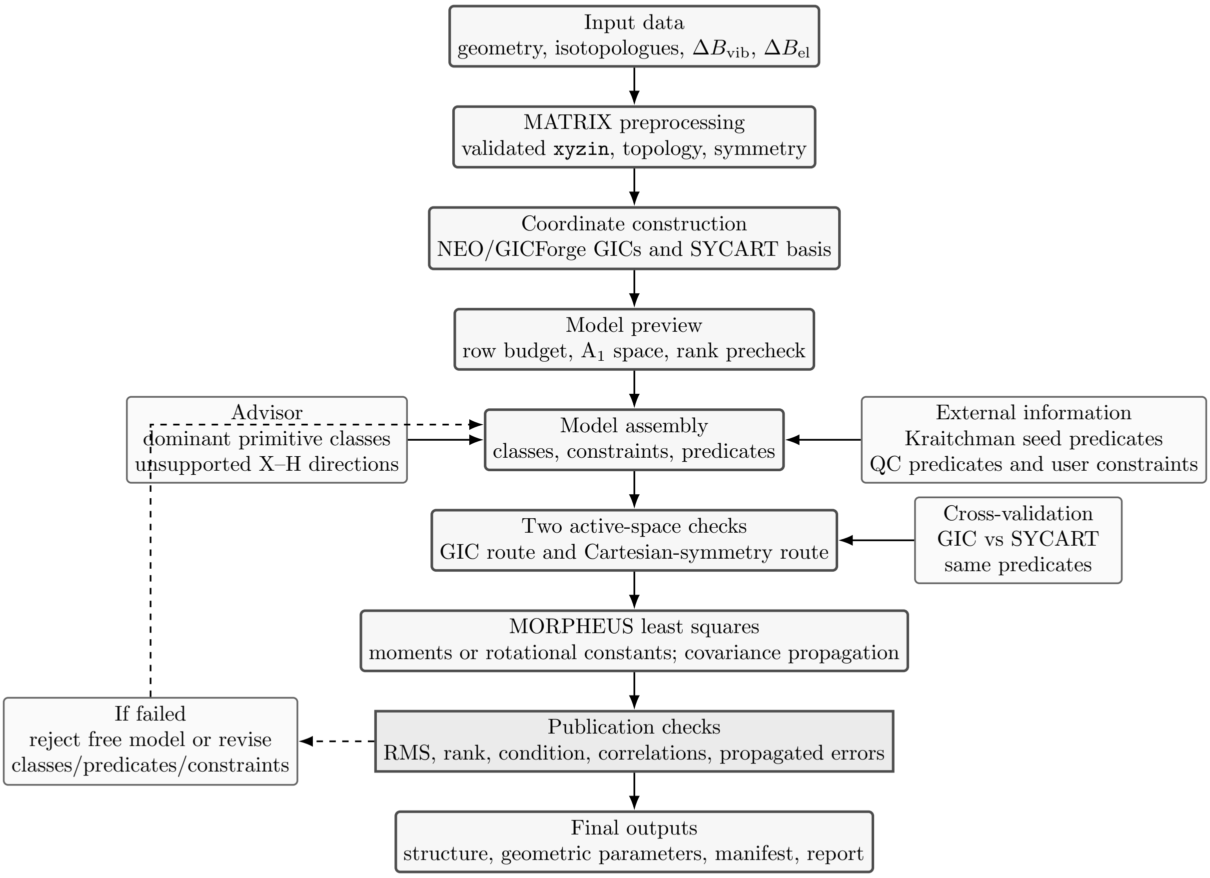

Figure 1: Suggested MORPHEUS workflow for semiexperimental structure refinement. The central vertical path is the reproducible pipeline; side boxes supply model-building information or validation. Kraitchman coordinates enter as experimental seed predicates, not as exact constraints.

possible bias from a particular internal-coordinate reduction. It is less transparent chemically: primitive constraints and predicates can still be projected onto the Cartesian basis, but the fitted parameters themselves are Cartesian symmetry displacements, not local structural variables.

2.3 Expected equivalence

For a well-posed least-squares model, the two active spaces should give the same final observable residuals and the same structural corrections within numerical tolerance. Differences are meaningful diagnostics. A large difference indicates an ill-posed model, an incomplete constraint projection, a singular coordinate block, or a mismatch between chemical priors and experimental information.

3 Model choices

3.1 Fully free refinement

The fully free model is the least biased option, but it is rarely meaningful unless the number and distribution of isotopologues are sufficient. MORPHEUS now rejects an underdetermined free refinement before iteration. This is essential: a formally small rotational RMS is not a structural result if unsupported directions are present.

3.2 Hard constraints

Hard constraints remove directions from the active least-squares space. They are appropriate when a coordinate is known not to be determined by the data, for example unsubstituted hydrogen local frames in a heavy-atom isotopic dataset. Hard constraints must be reported because they have zero variance along the removed directions.

3.3 Parameter classes

Parameter classes tie several corrections to one fitted variable. They do not force the final primitive values to be identical; they force their corrections from the reference structure to share one amplitude. Classes are useful when the data support a family correction but not each member separately. Examples are CC stretches, CO stretches, chemically related valence angles, or terminal CH_2 stretches.

3.4 Predicates

Predicates are weighted structural observations, not exact constraints. They are the preferred way to use high-quality reference structures when the rotational data alone do not determine all directions. A distance predicate with $\sigma = 0.003$ Å and an angle predicate with $\sigma = 0.3°$ mean that deviations are allowed but penalized according to the stated uncertainty. The covariance matrix includes the predicate weights, so final structural uncertainties are propagated rather than estimated by Monte Carlo sampling.

3.5 Kraitchman-derived predicates

Single-substitution Kraitchman coordinates provide an independent experimental geometric seed. MORPHEUS writes both a Kraitchman comparison table and, when possible, a principal-axis seed geometry. This seed can now be converted into local primitive predicates. The conservative default generates a predicate only when every atom in the primitive coordinate has a Kraitchman target; an optional partial mode also allows mixed primitives containing Kraitchman-seeded atoms and reference-geometry atoms. The resulting predicates are labelled as `kraitchman_seed_distance` or `kraitchman_seed_angle`. They should normally be assigned broader uncertainties than high-level QC predicates, because Kraitchman coordinates contain sign choices, small-coordinate instabilities, substitution assumptions, and vibration/electronic correction dependencies.

3.6 Automatic advisor

The advisor now builds disjoint suggestions from dominant primitive contributions to each GIC. A GIC is assigned to a chemical class only when one primitive family is dominant; classes are never allowed to mix stretches, bends, torsions, or other coordinate types. Suggested class names are chemically explicit, for example `CC_stretches`, `CO_stretches`, `CCO_bends`, or `HO_stretches`. When hydrogen substitutions are absent, dominant X–H angle directions may be proposed as fixed directions. The advisor is deliberately a model builder, not a black-box guarantee: its output is validated by rank, condition number, residuals, propagated uncertainties, and warning diagnostics.

4 Current numerical checks

The following tests use the fructose and deoxy-D-ribose datasets already converted to MATRIX `xyzin`. All fits use moments of inertia and the ABC components.

Advisor-only and X–H-blocked tests are useful diagnostics but are not reported as structural results here. They can give small rotational residuals while leaving geometric uncertainties too large, especially for valence angles and torsions. MORPHEUS therefore treats rank, condition number, covariance propagation, and high-correlation diagnostics as model checks rather than as replacements for chemically meaningful geometric uncertainties. The publishable models below use weighted structural predicates and pass the propagated-error checks.

With predicates the two active spaces become numerically equivalent. The small conditioning advantage of Cartesian symmetry coordinates in these two tests does not remove the need for GICs,

Table 1: GIC versus Cartesian symmetry coordinates with the same predicate model ($\sigma_R = 0.003$ Å, $\sigma_A = 0.3°$, $\sigma_D = 0.5°$).

System	Coordinate model	Parameters	Rank	RMS/MHz	Condition	max σ_R/Å	max σ_A/deg
Fructose	GIC	30	30	0.016560945	94.1	0.000301	0.030
Fructose	Cartesian symmetry	30	30	0.016560945	78.0	–	–
Deoxy-D-ribose	GIC	21	21	0.011124743	60.4	0.000559	0.055
Deoxy-D-ribose	Cartesian symmetry	21	21	0.011124743	54.0	–	–

because the chemical model, the constraints, and the final published parameters are naturally internal-coordinate quantities.

5 Recommended workflow

1. Build topology, symmetry, and non-redundant GICs once from the validated `xyzin`.
2. Preview the model: count rows, active coordinates, suggested classes, and fixed hydrogen directions.
3. Reject fully free models when the row budget or rank is insufficient.
4. Generate Kraitchman diagnostics and test Kraitchman predicates when isotope coverage is suitable.
5. Try advisor-generated classes and inspect rank, condition number, propagated errors, and high-correlation diagnostics.
6. Add hard constraints only for directions that are physically unsupported by the data.
7. Add predicates for chemically meaningful reference information, with documented uncertainties.
8. Run both GIC and Cartesian symmetry active spaces for final validation when the system is important enough for publication.

6 Final geometric parameters

Figure 1 and the diagnostics above define the workflow, but the structural result is the set of fitted geometric parameters and their propagated uncertainties. The following tables report the final GIC-based predicate refinements. C–H coordinates are omitted because they are not determined by the present isotopic datasets; O–H coordinates are retained and marked by their zero model variance when fixed by the hydrogen-frame constraint. Ring conformations are reported through Cremer–Pople-like Q and ϕ coordinates rather than by listing all intracyclic torsions.

6.1 Fructose: final geometric parameters

Coordinate	Atoms	Unit	Final value
R(1,11)	C–O	Å	1.41844 ± 3.0e-04
R(1,2)	C–C	Å	1.51892 ± 2.9e-04
R(10,21)	O–H	Å	0.96268 ± 0.0000
R(11,13)	O–H	Å	0.96166 ± 0.0000
R(12,24)	O–H	Å	0.96616 ± 0.0000
R(2,12)	C–O	Å	1.40861 ± 2.9e-04
R(2,3)	C–C	Å	1.52438 ± 2.8e-04
R(2,7)	C–O	Å	1.40764 ± 2.9e-04
R(3,4)	C–C	Å	1.51998 ± 2.9e-04
R(3,8)	C–O	Å	1.41741 ± 3.0e-04
R(4,5)	C–C	Å	1.51817 ± 2.9e-04
R(4,9)	C–O	Å	1.41636 ± 3.0e-04
R(5,10)	C–O	Å	1.41247 ± 3.0e-04
R(5,6)	C–C	Å	1.51383 ± 2.9e-04
R(6,7)	C–O	Å	1.42288 ± 2.9e-04
R(8,14)	O–H	Å	0.96415 ± 0.0000
R(9,15)	O–H	Å	0.96227 ± 0.0000
A(1,11,13)	C–O–H	deg	106.745 ± 0.000
A(1,2,12)	C–C–O	deg	109.660 ± 0.027
A(1,2,3)	C–C–C	deg	113.219 ± 0.026
A(1,2,7)	C–C–O	deg	104.871 ± 0.028
A(2,1,11)	C–C–O	deg	109.162 ± 0.028
A(2,12,24)	C–O–H	deg	105.741 ± 0.000
A(2,3,4)	C–C–C	deg	110.336 ± 0.022
A(2,3,8)	C–C–O	deg	111.817 ± 0.030
A(2,7,6)	C–O–C	deg	114.620 ± 0.024
A(3,2,12)	C–C–O	deg	106.577 ± 0.027
A(3,2,7)	C–C–O	deg	110.551 ± 0.022
A(3,4,5)	C–C–C	deg	110.624 ± 0.023
A(3,4,9)	C–C–O	deg	110.420 ± 0.030
A(3,8,14)	C–O–H	deg	106.023 ± 0.000
A(4,3,8)	C–C–O	deg	110.001 ± 0.030
A(4,5,10)	C–C–O	deg	110.801 ± 0.027
A(4,5,6)	C–C–C	deg	109.079 ± 0.023
A(4,9,15)	C–O–H	deg	106.847 ± 0.000
A(5,10,21)	C–O–H	deg	106.245 ± 0.000
A(5,4,9)	C–C–O	deg	107.759 ± 0.030
A(5,6,7)	C–C–O	deg	112.280 ± 0.024
A(6,5,10)	C–C–O	deg	108.932 ± 0.030
A(7,2,12)	O–C–O	deg	112.075 ± 0.028
QPck0001	C–C–C–C–C–O	rad	0.06526 ± 0.0008
PhiP0001	C–C–C–C–C–O	deg	-94.707 ± 0.690

6.2 Deoxy-D-ribose: final geometric parameters

Coordinate	Atoms	Unit	Final value
R(1,14)	C–O	Å	1.40512 ± 5.5e-04
R(1,2)	C–C	Å	1.51723 ± 5.4e-04
R(1,6)	C–O	Å	1.41593 ± 5.4e-04
R(14,17)	O–H	Å	0.96041 ± 0.0000
R(15,18)	O–H	Å	0.96295 ± 0.0000
R(16,19)	O–H	Å	0.96184 ± 0.0000
R(2,3)	C–C	Å	1.52476 ± 5.2e-04
R(3,15)	C–O	Å	1.41232 ± 5.6e-04
R(3,4)	C–C	Å	1.52472 ± 5.2e-04
R(4,16)	C–O	Å	1.42460 ± 5.4e-04
R(4,5)	C–C	Å	1.51347 ± 5.5e-04
R(5,6)	C–O	Å	1.42712 ± 5.4e-04
A(1,14,17)	C–O–H	deg	107.896 ± 0.000
A(1,2,3)	C–C–C	deg	111.677 ± 0.040
A(1,6,5)	C–O–C	deg	113.071 ± 0.045
A(2,1,14)	C–C–O	deg	107.723 ± 0.049
A(2,1,6)	C–C–O	deg	111.703 ± 0.044
A(2,3,15)	C–C–O	deg	111.373 ± 0.055
A(2,3,4)	C–C–C	deg	109.895 ± 0.043
A(3,15,18)	C–O–H	deg	106.121 ± 0.000
A(3,4,16)	C–C–O	deg	109.654 ± 0.048
A(3,4,5)	C–C–C	deg	109.931 ± 0.044
A(4,16,19)	C–O–H	deg	107.054 ± 0.000
A(4,3,15)	C–C–O	deg	110.117 ± 0.047
A(4,5,6)	C–C–O	deg	109.907 ± 0.043
A(5,4,16)	C–C–O	deg	110.690 ± 0.053
A(6,1,14)	O–C–O	deg	111.583 ± 0.052
QPck0001	C–C–C–C–C–O	rad	0.18371 ± 0.0015
PhiP0001	C–C–C–C–C–O	deg	-54.055 ± 0.474

7 Open developments

The next improvements should make the advisor more chemically aware without making it opaque. Useful extensions are automatic predicate-width selection from reference level, synthon-aware class splitting, explicit data-budget optimization, and reporting of why each suggested class was accepted or rejected. The final paper should also include examples where the advisor chooses different models as isotopic coverage is increased, because that directly demonstrates that MORPHEUS formalizes model construction rather than merely automating a fit.